D0 · Start-of-year diagnostic test

S1 · 30 min · no calculator · not marked. Give a brief justification. For any unsolved question, note what blocks you.

1. Factorise $x^2 - 5x + 6$, then solve $x^2 - 5x + 6 \leq 0$.

2. Differentiate $(x+1)e^x$ and determine the sign of the derivative.

3. $u_0 = 3$, $u_{n+1} = 2u_n$. Give u_3 and a formula for u_n.

4. Calculate $1 + 2 + \cdots + 20$ and $1 + 2 + 4 + 8 + 16$.

5. $A(1,2)$, $B(3,5)$, $C(5,8)$. Are the points collinear?

6. $P(A) = 0{,}4$, $P_A(B) = 0{,}5$, $P_{\overline{A}}(B) = 0.2$. Calculate $P(B)$.

7. A fair coin is tossed twice independently. Calculate the probability of getting exactly one head.

8. Is the statement "if $x^2 = 4$, then $x = 2$" true? Give the exact correction.

Solutions D0 and follow-up

1. $x^2 - 5x + 6 = (x-2)(x-3)$. The product is positive before 2 and after 3, negative between the two roots and zero at the roots. The required set is $[2,3]$.

2. By the product rule, the derivative is $e^x + (x+1)e^x = (x+2)e^x$. Since $e^x > 0$, it is negative for $x < -2$, zero at -2, positive for $x > -2$.

3. The sequence is geometric with common ratio 2: $u_n = 3 \times 2^n$ & $u_3 = 3 \times 8 = 24$.

4. The arithmetic sum is $\frac{20(1+20)}{2} = 210$. The geometric sum is $\frac{2^5-1}{2-1} = 31$.

5. $\overrightarrow{AB} = (2,3)$ and $\overrightarrow{AC} = (4,6) = 2\overrightarrow{AB}$. Since $A \neq B$, the points are collinear.

6. By total probability, $P(B) = 0{,}4 \times 0{,}5 + 0{,}6 \times 0{,}2 = 0{,}32$.

7. The favourable outcomes are heads-tails and tails-heads. Independence gives each the probability $\frac{1}{4}$; their union has probability $\frac{1}{2}$.

8. The claim is false: $x = -2$ is a counterexample. The exact equivalence is $x^2 = 4 \iff$ $x = 2$ or $x = -2$.

Revisit algebra and signs if questions 1-2 fail; sequences and indices for 3-4; coordinates for 5; reading events for 6-7; logic for 8. This diagnostic is used to choose the reviews, without permanently labelling the students.